

C01 AT1 APPLICATION BASED

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Question 1: Caesar Cipher

Scenario

A college event committee plans to share the meeting schedule only with its organizing members. To prevent unauthorized access, the committee decides to use the Caesar Cipher with a fixed shift value of 4.

The original message is:

Plaintext: "MEET AT NOON"

Based on this scenario, answer the following:

1. Encrypt the plaintext "MEET AT NOON" using a Caesar Cipher with a shift value of 4.
2. Illustrate the encryption process by showing how each letter is transformed into the corresponding ciphertext letter.
3. Explain how an authorized member can recover the original message using the same key.
4. Discuss one advantage and two limitations of the Caesar Cipher in securing confidential messages.

ANSWER:

Given:

- **Plaintext:** MEET AT NOON
- **Shift Value:** 4

1. Encrypt the Plaintext

In the Caesar Cipher, each letter is shifted 4 positions forward in the English alphabet.

Plaintext Letter M E E T A T N O O N

Shift (+4) Q I I X E X R S S R

Ciphertext: QIIX EX RSSR

2. Encryption Process

The Caesar Cipher replaces each letter with another letter that is **4 positions ahead** in the alphabet.

Examples:

- $M \rightarrow N \rightarrow O \rightarrow P \rightarrow \mathbf{Q}$
- $E \rightarrow F \rightarrow G \rightarrow H \rightarrow \mathbf{I}$
- $T \rightarrow U \rightarrow V \rightarrow W \rightarrow \mathbf{X}$
- $A \rightarrow B \rightarrow C \rightarrow D \rightarrow \mathbf{E}$
- $N \rightarrow O \rightarrow P \rightarrow Q \rightarrow \mathbf{R}$
- $O \rightarrow P \rightarrow Q \rightarrow R \rightarrow \mathbf{S}$

Spaces are **not encrypted**.

Therefore,

MEET AT NOON ----- QIIX EX RSSR

3. Decryption Process

The receiver knows that the message was encrypted using a **shift value of 4**.

To recover the original message, each ciphertext letter is shifted **4 positions backward**.

Ciphertext Q I I X E X R S S R

Shift (-4) M E E T A T N O O N

Recovered Plaintext: **MEET AT NOON**

4. Advantages and Limitations

Advantage

- The Caesar Cipher is simple to understand and easy to implement, making it suitable for learning the basic concepts of encryption.

Limitations

1. It has only **25 possible shift values**, so an attacker can easily try every shift using a brute-force attack.
2. It uses the **same shift for every letter**, making it vulnerable to frequency analysis and unsuitable for protecting confidential information.

Final Answer

- **Plaintext:** MEET AT NOON
- **Shift Value:** 4
- **Ciphertext:** QIIX EX RSSR
- **Decrypted Message:** MEET AT NOON

Question 2: Monoalphabetic Substitution Cipher (10 Marks)

Scenario

During a routine cybersecurity audit, a network analyst discovers an encrypted message while monitoring communication between two systems. The intercepted message is:

"GSRH RH Z HVXIVG"

Initial analysis indicates that the message was encrypted using a **monoalphabetic substitution cipher**, where each letter is consistently replaced by another letter according to a fixed substitution rule.

Based on this information, answer the following questions.

Questions

1. Determine the most likely original plaintext corresponding to the intercepted ciphertext.
2. Identify the classical encryption technique used to encrypt the message.

3. Explain the decryption process by showing how each ciphertext letter is converted back to the original plaintext.
4. Justify your answer by analyzing the word pattern, repeating letters, and characteristics of the substitution cipher.
5. Mention one advantage and one limitation of this encryption technique.

Answer :

Given Ciphertext: GSRH RH Z HVXIVG

1. Determine the Original Plaintext

The ciphertext is:

GSRH RH Z HVXIVG

By applying the correct substitution rule, the original plaintext is: **THIS IS A SECRET**

2. Identify the Encryption Technique

The message is encrypted using the Atbash Cipher, which is a type of monoalphabetic substitution cipher.

In the Atbash Cipher, each letter is replaced by its opposite letter in the alphabet.

Plain Letter A B C D E F G H I J K L M

Cipher Letter Z Y X W V U T S R Q P O N

Plain Letter N O P Q R S T U V W X Y Z

Cipher Letter M L K J I H G F E D C B A

3. Step-by-Step Decryption

Word 1: GSRH

Cipher G S R H

Plain T H I S

Result: THIS

Word 2: RH

Cipher R H

Plain I S

Result: **IS**

Word 3: Z

Cipher Z

Plain A

Result: **A**

Word 4: HVXIVG

Cipher H V X I V G

Plain S E C R E T

Result: **SECRET**

4. Final Plaintext

Combining all the words: **THIS IS A SECRET**

5. Justification

The original message is identified by analyzing the following characteristics:

- The ciphertext contains **four words** with the pattern **4–2–1–6**, which matches the common English phrase **"THIS IS A SECRET."**
- The two-letter word **"RH"** is likely to represent a common word such as **"IS."**
- The single-letter word **"Z"** naturally maps to **"A"** in the Atbash cipher.
- Every letter follows the reverse alphabet substitution rule consistently ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, ...), confirming that the message uses the **Atbash monoalphabetic substitution cipher**.
- The decoded sentence is meaningful and grammatically correct, confirming that **"THIS IS A SECRET"** is the most probable original message.

Final Answer

- **Ciphertext:** GSRH RH Z HVXIVG
- **Encryption Technique:** Atbash Cipher (Monoalphabetic Substitution Cipher)
- **Original Plaintext:** THIS IS A SECRET

Here is a **different scenario and question** (same concept, different wording) suitable for a university exam.

Question 3: Vigenère Cipher

Scenario

A software company uses the **Vigenère Cipher** to protect sensitive information exchanged between employees. Before sending a confidential message, an employee encrypts the word **"HELLO"** using the keyword **"KEY"**. The keyword is repeated until it matches the length of the plaintext, and each letter of the plaintext is encrypted using the corresponding letter of the keyword.

Based on this scenario, answer the following questions.

Questions

1. Encrypt the plaintext **"HELLO"** using the keyword **"KEY"**.
2. Show the complete encryption process by converting the plaintext and keyword into their alphabetical values and calculating the ciphertext.
3. Explain why the keyword is repeated during the encryption process.
4. How does the use of a repeating keyword make the Vigenère Cipher more secure than the Caesar Cipher?
5. Mention one advantage and one limitation of the Vigenère Cipher.

Answer :

Given:

- **Plaintext:** HELLO

- **Keyword:** KEY

1. Encrypt the Plaintext

The Vigenère Cipher encrypts each letter of the plaintext using the corresponding letter of the repeating keyword.

Step 1: Repeat the Keyword

Since the plaintext has 5 letters, repeat the keyword until it matches the length of the plaintext.

Plaintext H E L L O

Keyword K E Y K E

Step 2: Convert Letters to Numbers

Assign values to the letters (A = 0, B = 1, ..., Z = 25).

Plaintext H E L L O

Value 7 4 11 11 14

Keyword K E Y K E

Value 10 4 24 10 4

Step 3: Apply the Encryption Formula

Formula:

Ciphertext = (Plaintext + Keyword) mod 26

2. Ciphertext

Plaintext Value	Keyword Value	Calculation	Cipher Value	Cipher Letter
7	10	$(7 + 10) = 17$	17	R
4	4	$(4 + 4) = 8$	8	I
11	24	$(11 + 24) = 35$ $\rightarrow 35 \bmod 26 = 9$	9	J

11	10	$(11 + 10) = 21$	21	V
14	4	$(14 + 4) = 18$	18	S

The encrypted message is:

RIJVS

3. Step-by-Step Encryption Process

- Write the plaintext: **HELLO**
- Repeat the keyword: **KEYKE**
- Convert each letter into its numerical value.
- Add the plaintext value and keyword value.
- If the sum is greater than 25, take **modulo 26**.
- Convert the resulting numbers back into letters.

Therefore,

HELLO-----RIJVS

4. How the Repeating Keyword Influences the Ciphertext

The keyword is repeated (**KEYKE**) so that every plaintext letter has a corresponding key letter.

Plaintext H E L L O

Keyword K E Y K E

Shift +10 +4 +24 +10 +4

Ciphertext R I J V S

Explanation:

- Each letter of the keyword provides a different shift value.

- The keyword repeats until it matches the length of the plaintext.
- Since different shifts are applied to different letters, repeated plaintext letters may produce different ciphertext letters.
- For example, the two **L** letters are encrypted differently:
 - First **L + Y** → **J**
 - Second **L + K** → **V**
- This makes the Vigenère Cipher more secure than the Caesar Cipher, which uses only one fixed shift for all letters.

5. Advantages and Limitations

Advantage

- The Vigenère Cipher uses multiple shift values based on a keyword, making it more resistant to simple frequency analysis than the Caesar Cipher.

Limitation

- If the keyword is short or repeated frequently, attackers can analyze patterns and potentially break the cipher using cryptanalysis techniques such as the Kasiski examination.

Final Answer

- **Plaintext:** HELLO
- **Keyword:** KEY
- **Repeated Keyword:** KEYKE
- **Ciphertext:** RIJVS

Question 4: Steganography

Scenario

A digital forensic team is investigating a suspected data leakage incident in an organization. During the investigation, they find that a company logo image may have been used to secretly

transmit confidential information. Although the image appears normal, investigators suspect that hidden data has been embedded using a steganographic technique.

Based on this scenario, answer the following questions:

1. Explain how investigators can detect whether hidden data is present inside the image.
2. Describe two methods used in digital forensics to analyze images for hidden information.
3. Recommend two techniques that can improve the security of hidden communications against detection or unauthorized extraction.
4. Differentiate between steganography and encryption with suitable examples.

Answer :

1. Detecting Hidden Data in an Image

Investigators can determine whether an image contains hidden information using the following methods:

Method 1: Steganalysis Tools

Specialized software such as **StegExpose**, **OpenStego**, or **zsteg** scans images to detect hidden messages. These tools analyze pixel values and identify unusual patterns that may indicate embedded data.

Method 2: Statistical Analysis

Investigators examine the image's **Least Significant Bit (LSB)** values, color distribution, and histogram. If the pixel values show unexpected changes, it may indicate that secret information has been hidden inside the image.

2. Digital Forensic Methods

a) LSB (Least Significant Bit) Analysis

Many steganography techniques hide data by modifying the least significant bit of image pixels. Investigators compare the LSB values to detect abnormal modifications.

b) Histogram Analysis

A histogram represents the distribution of colors in an image. Hidden data may slightly alter this distribution, helping investigators identify possible steganographic content.

3. Techniques to Improve Security

Technique 1: Encrypt the Message Before Hiding

Before embedding the message in the image, encrypt it using a strong encryption algorithm such as **AES (Advanced Encryption Standard)**.

Explanation:

Even if an attacker extracts the hidden data, the information remains unreadable without the correct decryption key.

Technique 2: Randomized or Key-Based Embedding

Instead of storing data in consecutive pixels, use a secret key to randomly select pixel locations for embedding the hidden message.

Explanation:

Random placement makes it much more difficult for attackers and steganalysis tools to locate or extract the hidden information.

4. Difference Between Steganography and Encryption

Steganography	Encryption
Hides the existence of the message.	Converts the message into unreadable text.
Uses images, audio, or video files to conceal data.	Uses encryption algorithms and keys.
Example: Hiding text inside an image.	Example: Encrypting a file using AES.

Conclusion

Digital forensic investigators use **steganalysis tools**, **LSB analysis**, and **histogram analysis** to detect hidden information inside images. To improve the security of hidden communications, organizations should **encrypt the message before embedding it** and **use randomized key-based embedding**, making detection and extraction much more difficult.

Question 5: Rail Fence Cipher

Scenario

A company continues to use a **3-Rail Rail Fence Cipher** to secure internal communication in its legacy software system. During a security audit, the administrator receives the following encrypted message:

Ciphertext: WECRLTEERDSOEFEAOCAIVDEN

Based on this scenario, answer the following questions:

1. Decrypt the given ciphertext and determine the original plaintext message.
2. Explain the decryption process of the 3-Rail Rail Fence Cipher.
3. Discuss why the Rail Fence Cipher is considered insecure in modern communication systems.
4. Suggest one modern encryption technique that can replace this method and explain why it is more secure.

Answer :

1. Determine the Original Plaintext

Given Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

The message is encrypted using a **3-Rail Rail Fence Cipher**.

After reversing the encryption process, the original plaintext is:

WE ARE DISCOVERED FLEE AT ONCE

(Without spaces: **WEAREDISCOVEREDFLEEATONCE**)

2. Decryption Process

The Rail Fence Cipher is a **transposition cipher**, where letters are **rearranged** rather than replaced.

Step 1: Identify the number of rails.

- Number of Rails = **3**

Step 2: Reconstruct the zigzag pattern.

Rail 1 : W . . . E . . . C . . . R . . . L . . . T . . . E

Rail 2 : . E . R . D . S . O . E . E . F . E . A . O . C .

Rail 3 : . . C . . . A . . . I . . . V . . . D . . . E . . N

Step 3: Read the letters in the zigzag order.

WEAREDISCOVEREDFLEEATONCE

Therefore, the original message is:

WE ARE DISCOVERED FLEE AT ONCE

3. Why is the Rail Fence Cipher Vulnerable?

The Rail Fence Cipher is not suitable for modern communication because:

a) It only rearranges letters

- The letters themselves are not changed.
- The frequency of letters remains the same, making cryptanalysis easier.

b) Small Key Space

- Only a few rail values (2, 3, 4, etc.) are possible.
- Attackers can try all rail combinations using brute-force methods.

c) Easy to Break

- Modern computers can quickly test different rail arrangements and recover the original message within seconds.

4. Practical Enhancement

Use AES (Advanced Encryption Standard)

AES is a modern symmetric encryption algorithm that provides much stronger security than the Rail Fence Cipher.

Advantages of AES

- Uses **128-bit, 192-bit, or 256-bit encryption keys.**

- Resistant to brute-force attacks.
- Encrypts the actual data instead of simply rearranging letters.
- Widely used for secure communication in banking, cloud services, and government applications.

Conclusion

The ciphertext "**WECRLTEERDSOEFEAOCAIVDEN**" decrypts to:

WE ARE DISCOVERED FLEE AT ONCE

Although the Rail Fence Cipher is simple to understand, it offers very limited security because it only rearranges the characters. Replacing it with a modern encryption algorithm such as **AES** significantly improves the confidentiality and security of sensitive information.